

Structured DNN-Based Receiver for Millimeter-Wave MIMO With Nonlinear Distortions

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Abstract—This work deals with the combined effect of nonlinear distortions and inter-channel interference in millimeter wave multi-input multi-output (MIMO) communications. Deep neural networks (DNNs) can be used to handle the effect, but they often require a large number of pilot symbols, hindering their applications. With the aim of online training using a relatively small number of pilot symbols, we design a deep neural network (DNN) architecture carefully, which consists of a fully connected linear hidden layer and a non-fully connected nonlinear hidden layer. The linear hidden layer is used to deal with the co-channel interference and the nonlinear hidden layer is used to handle the nonlinear distortions. Moreover, the parameters of the DNN are properly tied to reduce the number of independent parameters. With such a DNN, the receiver is much efficient in terms of training overhead and symbol error rate performance, compared to conventional (DNN-based) techniques. Simulation results demonstrate the superiority of the proposed DNN-based receiver.

Index Terms—Nonlinearity, MIMO, neural networks, DNN, millimeter wave, power amplifier.

I. INTRODUCTION

MULTI-INPUT multi-output (MIMO) communications in the millimeter wave (mmWave) frequency band are a promising technology to meet the ever growing data rate requirements for the 5G wireless networks and beyond [1]. As the mmWave signals suffer from high attenuation, high transmission power is usually required [2], and the mmWave power amplifiers (PAs) show nonlinear characteristics [3]. To achieve high data rate, high order modulations are required, making the communications vulnerable to nonlinear distortions. This leads to combined effect of nonlinear distortions and co-channel interference, which needs to be handled properly.

To handle the nonlinear distortions, Volterra-series based technique is proposed to linearize or compensate the PA nonlinearity [4]. However, Volterra-series requires a large

number of coefficients, which hinders its applications. Then memory polynomial based techniques are proposed with simplified structure to address the PA nonlinearity [5]. However, these polynomial based techniques suffer from numerical instability in determining their coefficients especially in the case of high order polynomials [6]. Neural networks (NNs) are an alternative promising technique to model and mitigate the PA nonlinearity due to their universal approximation capability [7]–[11]. Single-hidden-layer neural networks are proposed to replace the “look-up-table” method for PA predistortion in [7]. Deep NNs (DNNs) are also investigated, e.g., [12], where various number of layers and different activation functions are studied for NN-based PA predistortion. In [9], hyperparameter optimization is investigated in a fully connected two-hidden-layer neural network for PA nonlinearity modeling.

Although mitigation of either nonlinear distortion or co-channel interference has been extensively studied, there are few investigations on the dealing with the combined effect of nonlinear distortion and co-channel interference. In this case, the techniques used to deal with them separately are often not applicable as the nonlinear distortion and co-channel inference are unknown and tangled. A potential way is to use DNN to handle them at the same time, which, however, will lead to an excessive number of parameters, often requiring an impractically large number of pilot symbols in training.

In this letter, we aim to design a DNN that can be trained online with a relatively small number of pilot symbols. To achieve this, we design the architecture of the DNN carefully, and restrict the number of layers and parameters. Specifically, by considering the cascade of nonlinear distortion and linear co-channel inference, we design a DNN consisting of a fully-connected linear hidden layer, followed by a non-fully connected nonlinear hidden layer. The first hidden layer is used to deal with the linear co-channel interference, and the second hidden layer is used to deal with the PA nonlinear distortions. Moreover the parameters of the DNN are properly tied to significantly reduce the number of parameters. We show that such a special DNN is very efficient to mitigate the combined effects. Simulation results demonstrate the superiority of the proposed DNN compared to existing receivers.

Notations: Boldface lower-case and upper-case letters denote vectors and matrices. The superscripts $(\cdot)^T$ and $(\cdot)^H$ represent the transpose and conjugate transpose operations. We use $|x|$ to denote the amplitude of x . $\Re\{\cdot\}$ and $\Im\{\cdot\}$ represent the real and imaginary parts of a complex number.

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II. SYSTEM MODEL AND CONVENTIONAL RECEIVERS

A. System Model

Consider a MIMO system with K transmit antennas and N receive antennas equipped at a receiver. The K transmit antennas can be equipped at a single transmitter in the case of point-to-point transmission or multiple transmitters in the case of multi-user uplink transmission.¹

The PA model regulated by IEEE 802.11ad task group (TG) [3] is used, which is characterized by the amplitude to amplitude conversion $A(|x|)$ and amplitude to phase conversion $\Phi(|x|)$, i.e.,

$$A(|x|) = \frac{\alpha_a |x|}{(1 + (\alpha_a \frac{|x|}{x_{\text{sat}}})^{2\sigma_a})^{\frac{1}{2\sigma_a}}}, \quad \Phi(|x|) = \frac{\alpha_\phi |x|^{q_1}}{1 + (\frac{|x|}{\beta_\phi})^{q_2}}, \quad (1)$$

where x is the signal input to the PA, and α_a , α_ϕ , β_ϕ , σ_a , x_{sat} , q_1 and q_2 are the relevant parameters. Then the distorted signal can be expressed as

$$s = f(x) = A(|x|)e^{j(\angle(x) + \Phi(|x|))}, \quad (2)$$

where $\angle(x)$ denotes the angle of the complex signal x . The received signal at time instant m is

$$\mathbf{y}(m) = \mathbf{H}\mathbf{s}(m) + \mathbf{n}(m) = \mathbf{H}f(\mathbf{x}(m)) + \mathbf{n}(m), \quad (3)$$

where $\mathbf{H}$ is an $N \times K$ mmWave MIMO channel matrix, $\mathbf{y}(m) = [y_1(m), y_2(m), \dots, y_N(m)]^T$, $\mathbf{s}(m) = f(\mathbf{x}(m))$ with $\mathbf{x}(m) = [x_1(m), x_2(m), \dots, x_K(m)]^T$ being the transmitted signals of the K antennas at time instant m , and $\mathbf{n}(m)$ denotes an additive white Gaussian noise vector. Note that $\mathbf{y}$, $\mathbf{H}$, $\mathbf{s}$ and $\mathbf{n}$ are all complex valued, and the model can be rewritten as a real-valued form as

$$\underbrace{\begin{bmatrix} \Re\{\mathbf{y}(m)\} \\ \Im\{\mathbf{y}(m)\} \end{bmatrix}}_{\mathbf{y}'} = \underbrace{\begin{bmatrix} \Re\{\mathbf{H}\} & -\Im\{\mathbf{H}\} \\ \Im\{\mathbf{H}\} & \Re\{\mathbf{H}\} \end{bmatrix}}_{\mathbf{H}'} \underbrace{\begin{bmatrix} \Re\{\mathbf{s}(m)\} \\ \Im\{\mathbf{s}(m)\} \end{bmatrix}}_{\mathbf{s}'} + \underbrace{\begin{bmatrix} \Re\{\mathbf{n}(m)\} \\ \Im\{\mathbf{n}(m)\} \end{bmatrix}}_{\mathbf{n}'}. \quad (4)$$

The aim is to train a receiver based on training symbols and thereby recover the transmitted data symbols based on the received signal. Suppose there are M_0 training samples $\{(\mathbf{p}(m), \mathbf{t}(m)), m = 1, \dots, M_0\}$, where $\mathbf{t}(m) = [t_1(m), t_2(m), \dots, t_K(m)]^T$ denotes the training symbols, and $\mathbf{p}(m) = [p_1(m), p_2(m), \dots, p_N(m)]^T$ denotes the corresponding received signal.

B. Zero Forcing (ZF) Receiver

The ZF receiver can be directly trained using the training samples. In this case the weight of the receiver for user k can be expressed as

$$\mathbf{w}_{k(\text{Train})}^{\text{ZF}} = (\mathbf{P}^H \mathbf{P})^{-1} \mathbf{P}^H \mathbf{t}_k, \quad (5)$$

where $\mathbf{P} = [\mathbf{p}(1), \mathbf{p}(2), \dots, \mathbf{p}(M)]^T$ is the matrix of the received signals and $\mathbf{t}_k = [t_k(1), t_k(2), \dots, t_k(M)]^T$ is the transmitted pilot signals for user k . It is noted that the ZF receiver is a linear one and it does not have dedicated mechanism to mitigate the nonlinear distortions.

¹The proposed receiver works for both cases, but in the simulations, only the latter is focused.

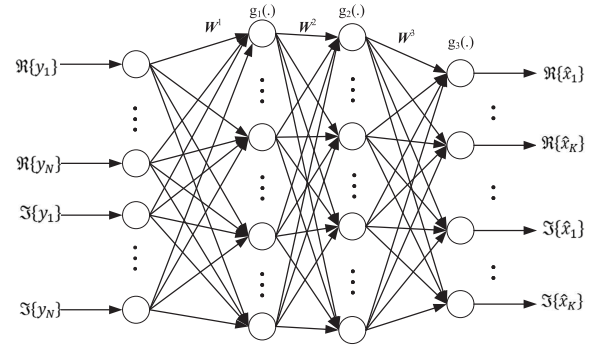


Fig. 1. Conventional DNN receiver (with two hidden layers).

C. Conventional DNN-Based Receiver

Real-valued two-hidden-layer neural networks are employed to deal with the PA nonlinearity [9], [13]. Fig. 1 shows the structure of a two-hidden-layer neural network.

With the real and imaginary parts of the training symbols separated, $\mathbf{p}'(m) = [\Re\{\mathbf{p}(m)\}^T, \Im\{\mathbf{p}(m)\}^T]^T$ is the input of the NN and the expected output is $\mathbf{t}'(m) = [\Re\{\mathbf{t}(m)\}^T, \Im\{\mathbf{t}(m)\}^T]^T$. With fully connected layers, the output of the i th neuron in layer $l + 1$ ($l = 0, 1$ and 2) is

$$z_i^{l+1} = g_{l+1}(\psi_i^{l+1}), i = 1, 2, \dots, D_{l+1}, \quad (6)$$

where

$$\psi_i^{l+1} = \sum_{j=1}^{D_l} w_{ij}^{l+1} z_j^l + b_i^{l+1} \quad (7)$$

is the i th input of layer $l + 1$, w_{ij}^{l+1} is the weight that connects the j th neuron in the previous layer l and the i th neuron in current layer $l + 1$, b_i is bias, $g_{l+1}(\cdot)$ is the element-wise activation function of neurons in layer $l + 1$, and D_l is the number of hidden nodes of previous layer. Note that z_i^0 is the i th input of the entire network and D_0 is input dimension of the network. Eq. (6) can be written as a vector form

$$\mathbf{z}^{l+1} = g_{l+1}(\Psi^{l+1}), \quad (8)$$

where

$$\Psi^{l+1} = \mathbf{W}^{l+1} \mathbf{z}^l + \mathbf{b}^{l+1}, \quad (9)$$

$\Psi^{l+1} = [\psi_1^{l+1}, \psi_2^{l+1}, \dots, \psi_{D_{l+1}}^{l+1}]^T$, $\mathbf{z}^{l+1} = [z_1^{l+1}, z_2^{l+1}, \dots, z_{D_{l+1}}^{l+1}]^T$, $\mathbf{b}^{l+1} = [b_1^{l+1}, b_2^{l+1}, \dots, b_{D_{l+1}}^{l+1}]^T$, and

$$\mathbf{W}^{l+1} = \begin{bmatrix} w_{11}^{l+1} & \dots & w_{1D_l}^{l+1} \\ \vdots & \ddots & \vdots \\ w_{D_{l+1}1}^{l+1} & \dots & w_{D_{l+1}D_l}^{l+1} \end{bmatrix}. \quad (10)$$

For regression, the activation function of the output layer is usually linear. Mathematically, the activation function is simply $\mathbf{z}^3 = \Psi^3$. The conventional DNN-based receiver involves a large number of parameters (i.e., $\mathbf{W}^1$, $\mathbf{W}^2$ and $\mathbf{W}^3$), which causes difficulties in training as a larger number of training symbols are required and the receiver is prone to overfitting. The computational cost can also be a concern.

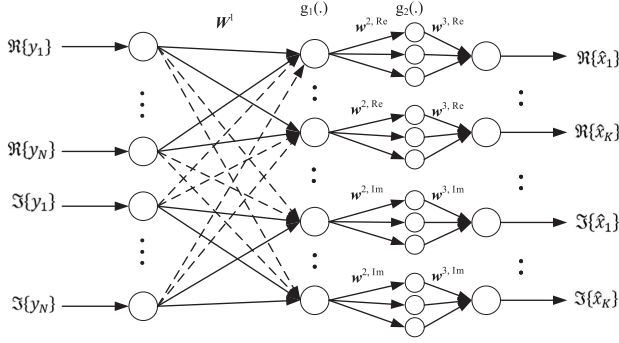


Fig. 2. Proposed structured DNN based receiver.

D. ZF Receiver With Perfect $\mathbf{H}$

Although it is unknown how to estimate the channel matrix $\mathbf{H}$ when the nonlinearity of PA is unknown, we include the ZF receiver with perfect $\mathbf{H}$ as a benchmark. The weight for transmit antenna k can be represented as

$$(\mathbf{w}_k^{\text{ZF}})^T = [(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H]_k^T, \quad (11)$$

where $[\cdot]_k$ denotes the k th row of the matrix. It can be seen that the receiver simply ignores the PA nonlinear distortion.

III. STRUCTURED DNN-BASED RECEIVER

To enable practical online training of DNN with limited pilot symbols, we propose to use a DNN shown in Fig. 2, which consists of a fully connected linear hidden layer and a non-fully connected nonlinear hidden layer. The structure of the NN is inspired by the system model (3), i.e., the transmitted symbols are first distorted by PAs and then undergo the co-channel interference. Intuitively, the receiver should perform corresponding operations to first mitigate the co-channel interference and then compensate the nonlinear distortions.

- The first hidden layer is used to deal with the co-channel interference. The activation function of the first hidden layer is linear as the co-channel interference is in a linear form. We note that the in-phase and quadrature-phase of the received signal are separated so that the DNN is real valued. Therefore, the number of hidden nodes of the first hidden layer is $2K$.
- At the outputs of the nodes in the first hidden layer, we assume that the interference-free signal is obtained but it is still nonlinearly distorted. Hence, the output of each hidden node is connected to a sub-NN with a single hidden layer (forming the second hidden layer of the DNN) to compensate the nonlinear distortions.

According to Fig. 2, the output of the first hidden layer is

$$\mathbf{z}^1(m) = \mathbf{W}^1 \mathbf{p}'(m) + \mathbf{b}^1, \quad (12)$$

where $\mathbf{p}'(m)$ is the m th training vector, and $\mathbf{W}^1$ is the weight matrix with size $2K \times 2N$. Considering the structure of $\mathbf{H}^1$ in (4), we tie the entries of the weight matrix $\mathbf{W}^1$, so that we have

$$\mathbf{W}^1 = \begin{bmatrix} \mathbf{W}^{11} & \mathbf{W}^{12} \\ -\mathbf{W}^{12} & \mathbf{W}^{11} \end{bmatrix}, \quad (13)$$

where $\mathbf{W}^{11}$ and $\mathbf{W}^{12}$ are matrices with dimension $K \times N$. It can be seen that the number of parameters is reduced from $4KN$ to $2KN$, which can be helpful to reduce the number of pilot symbols in DNN training.

We assume that the co-channel interference is mitigated through the first hidden layer, so the first half of $\mathbf{z}^1(m)$ corresponds to the in-phase components of the distorted signal $s(m)$ and the second half corresponds to the quadrature ones. As a NN with a single hidden layer has the property of universal approximation, a single hidden layer NN is sufficient to deal with the nonlinear distortions due to PAs. Hence the output of each hidden node is connected to a sub-NN with a single hidden layer, where the number of hidden nodes is N' . In this way, the second hidden layer is formed. From (1), the in-phase and quadrature-phase components suffer from different nonlinearity. So the parameters of the sub-NNs for the in-phase component are tied and those for the quadrature-phase components are tied. This reduces the number of parameters of sub-NNs from $6KN'$ to $6N'$, making the training process more efficient as shown later.

With the i th output z_i^1 from the first hidden layer as input, the corresponding output of the second hidden layer can be expressed as

$$\mathbf{z}_i^2(m) = \begin{cases} g_2(\mathbf{w}^{2,\text{Re}} z_i^1 + \mathbf{b}^{2,\text{Re}}) & \text{for } i = 1, \dots, K, \\ g_2(\mathbf{w}^{2,\text{Im}} z_i^1 + \mathbf{b}^{2,\text{Im}}) & \text{for } i = K + 1, \dots, 2K, \end{cases} \quad (14)$$

where $\mathbf{z}_i^2(m) = [z_{i1}^2(m), z_{i2}^2(m), \dots, z_{iN'}^2(m)]^T$ is the output of the i th sub-NN in the second hidden layer, $g_2(\cdot)$ is the activation function of the second hidden layer (which is chosen as Tanh), $\mathbf{w}^{2,\text{Re}} = [w_1^{2,\text{Re}}, w_2^{2,\text{Re}}, \dots, w_{N'}^{2,\text{Re}}]^T$ and $\mathbf{b}^{2,\text{Re}} = [b_1^{2,\text{Re}}, b_2^{2,\text{Re}}, \dots, b_{N'}^{2,\text{Re}}]^T$ are the input weights and biases of the second hidden layer tied for the prediction of the in-phase components of the transmitted signal $\mathbf{t}'(m)$, and similarly $\mathbf{w}^{2,\text{Im}}$ and $\mathbf{b}^{2,\text{Im}}$ are the parameters tied for the prediction of the quadrature-phase components.

The i th output of the entire network is expressed as

$$\mathbf{z}_i^3(m) = \begin{cases} (\mathbf{w}^{3,\text{Re}})^T \mathbf{z}_i^2(m) & \text{for } i = 1, \dots, K, \\ (\mathbf{w}^{3,\text{Im}})^T \mathbf{z}_i^2(m) & \text{for } i = K + 1, \dots, 2K, \end{cases} \quad (15)$$

where $\mathbf{w}^{3,\text{Re}} = [w_1^{3,\text{Re}}, w_2^{3,\text{Re}}, \dots, w_{N'}^{3,\text{Re}}]^T$ and $\mathbf{w}^{3,\text{Im}} = [w_1^{3,\text{Im}}, w_2^{3,\text{Im}}, \dots, w_{N'}^{3,\text{Im}}]^T$ are the output weights to obtain the estimate of the in-phase and quadrature-phase components of the transmitted signal $\mathbf{t}'(m)$. Once the activation function of the output layer is linear and the biases of the output layer are zero. To train the DNN, we minimize the cost function

$$\text{Loss} = \frac{1}{M_0} \sum_{m=1}^{M_0} E(m), \quad (16)$$

where $E(m) = \frac{1}{2K} \sum_{k=1}^{2K} E_k(m)$, $E_k(m) = (e_k(m))^2$, $e_k(m) = z_k^3(m) - t_k'(m)$ and M_0 is the training length. The cost function is calculated to update the weights $\{W_{ij}^l\}$ and biases $\{b_i^l\}$ using back propagation [14]. It is noted that the parameters of the DNN are tied, and some details about the parameter update are given in the Appendix. Once the DNN is trained, it is applied to the received signals $\mathbf{y}(m)$ to estimate the transmitted data symbols $\{x_k(m)\}$, and the estimated symbols are denoted by $\{\hat{x}_k(m)\}$. Then, we make a hard decision

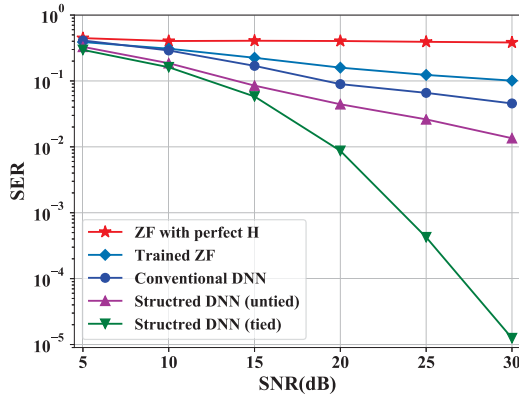


Fig. 3. SER of receivers trained with 1000 pilot symbols.

on $x_k(m)$ based on $\tilde{x}_k(m)$, i.e.,

$$\hat{x}_k(m) = \underset{c}{\operatorname{argmin}} |\tilde{x}_k(m) - c|, \quad (17)$$

where c belongs to the QAM symbol alphabet.

IV. SIMULATION RESULTS

Assume that the receiver (i.e., base station) is equipped with a uniform linear array of $N = 10$ antennas, and there are $K = 5$ users, each equipped with a single antenna. 16-QAM is used for the modulation. According to the Saleh-Valenzuela channel model [15], the channel vector $\mathbf{h}_k$ between the k th user and the N receive antennas can be written as

$$\mathbf{h}_k = \sqrt{\frac{N}{Q_k}} \sum_{q=1}^{Q_k} \beta_{kq} \mathbf{a}(\theta_{kq}), \quad (18)$$

where θ_{kq} is the incident angle of the q th path, $\mathbf{a}(\theta_{kq}) = \frac{1}{\sqrt{N}} [1, e^{-j2\pi d \sin(\theta_{kq})/\lambda}, \dots, e^{-j2\pi d \sin(\theta_{kq})(N-1)/\lambda}]^T$ is the length- N steering vector with antenna spacing d , λ is the wavelength of carrier, Q_k is the number of paths for user k , and β_{kq} is the complex gain of the q th path. The parameters are set as in [16], i.e., $d = \lambda/2$, $Q = 3$, β_{kq} follows Gaussian distribution with zero mean and unity variance, and θ_{kq} is uniformly drawn from $(-0.5\pi, 0.5\pi]$. As in [2], [3], the parameters used for the PA nonlinearity are $\alpha_a = 4.65$, $\alpha_\phi = 2560$, $\beta_\phi = 0.114$, $\sigma_a = 0.81$, $x_{\text{sat}} = 0.58$, $q_1 = 2.4$ and $q_2 = 2.3$. The SNR is defined as $\text{SNR} = P_x/\sigma_n^2$, where P_x is the power of the signal of a user, and σ_n^2 is the power of the noise (per receive antenna). The deep learning framework *Tensorflow* is used for the simulations. Batch gradient descent is used. Cross-validation is used to avoid overfitting and ensure the generality of the trained model. We use 80% and 20% of the dataset for training (including 3-fold validation) and testing. Through the validation data set, we determine that the batch size is 50 and the times of epoch is 300. Adam optimizer with a learning rate 0.01 is employed to update parameters. In the proposed structured DNN, $N' = 30$.

Figure 3 shows the symbol error rate (SER) of various receivers including the proposed DNN with tied and un-tied parameters, the conventional DNN based receiver and two conventional receivers (i.e., ZF with perfect $\mathbf{H}$ and trained ZF). The length of pilot symbols is 1000. It can be seen that the

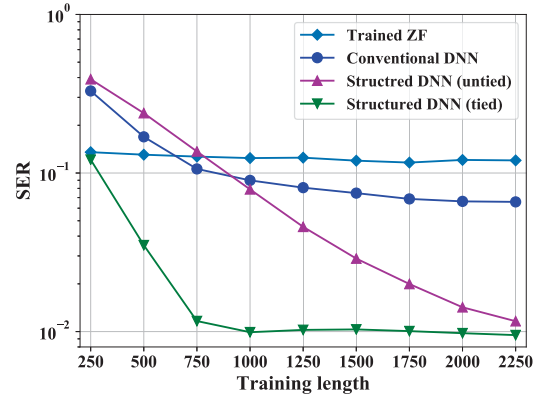


Fig. 4. SER of various receivers versus training length.

conventional ZF receiver does not work properly due to the lack of nonlinear distortion mitigation mechanism. The trained ZF performs better as the impact of nonlinearity is included in training, but the SER saturates at 10^{-1} due to its limited capability in handling nonlinear distortions. For the conventional DNN, the number of neurons in the first and second hidden layers are 40 and 50 respectively, and Tanh is used for both hidden layers. It has better performance than the ZF receivers but its performance is still not satisfactory. In contrast, the receiver using the proposed DNN with tied parameters outperforms other receivers significantly. We can also see that the performance of the proposed DNN with tied parameters is significantly better than that with untied parameters. This is because, with limited pilot symbols, the DNN with untied parameters cannot be well trained.

Figure 4 shows the SER performance of various receivers, which are trained with different numbers of pilot symbols. We assume SNR=20dB. The trained ZF receiver does not work properly even with more training symbols. The performance of the conventional DNN based receiver becomes better with the increase of training length, but the improvement is not significant. The proposed DNN with untied parameters achieves similar performance as the DNN with tied parameters when the training length is over 2000. This demonstrated that, with the parameters tied, more efficient training can be achieved.

Although the complexity of DNN based receivers is larger than that of the conventional ZF receiver, DNN-based receivers significantly outperform the ZF receiver. Compared to the conventional DNN based receiver, the structured DNN based receiver uses only two hidden layers and has a special architecture, leading to lower complexity. Moreover, the number of pilot symbols required by the structured DNN can be significantly reduced, which is crucial to practical applications.

APPENDIX

PARTIAL DERIVATIVES USED IN BACK-PROPAGATION

In the proposed DNN, the parameters are tied. We provide the partial derivatives used in back-propagation. According to the chain rule, the partial derivative with respect to $w_i^{3,\text{Re}}$ can be computed as

$$\frac{\partial E}{\partial w_i^{3,\text{Re}}} = \frac{1}{2K} \sum_{k=1}^K \frac{\partial E_k}{\partial z_k^3} \frac{\partial z_k^3}{\partial \psi_k^3} \frac{\partial \psi_k^3}{\partial w_i^{3,\text{Re}}}, \quad (19)$$

where

$$\frac{\partial E_k}{\partial z_k^3} = 2e_k, \quad \frac{\partial z_k^3}{\partial \psi_k^3} = 1, \quad \text{and} \quad \frac{\partial \psi_k^3}{\partial w_{i, \text{Re}}^3} = z_{ki}^2. \quad (20)$$

Thus, substituting (20) into (19), we have

$$\frac{\partial E}{\partial w_{i, \text{Re}}^3} = \frac{1}{K} \sum_{k=1}^K e_k z_{ki}^2, \quad (21)$$

and similarly the partial derivative with respect to $w_i^{3, \text{Im}}$ is

$$\frac{\partial E}{\partial w_{i, \text{Im}}^3} = \frac{1}{K} \sum_{k=K+1}^{2K} e_k z_{ki}^2. \quad (22)$$

The partial derivative with respect to $w_i^{2, \text{Re}}$ can be expressed as

$$\frac{\partial E}{\partial w_{i, \text{Re}}^2} = \frac{1}{2K} \sum_{k=1}^K \frac{\partial E_k}{\partial z_k^3} \frac{\partial z_k^3}{\partial \psi_k^3} \frac{\partial \psi_k^3}{\partial z_{ki}^2} \frac{\partial z_{ki}^2}{\partial \psi_{ki}^2} \frac{\partial \psi_{ki}^2}{\partial w_i^{2, \text{Re}}}, \quad (23)$$

where

$$\frac{\partial \psi_k^3}{\partial z_{ki}^2} = w_i^{3, \text{Re}}, \quad \frac{\partial \psi_{ki}^2}{\partial w_i^{2, \text{Re}}} = z_k^1, \quad \frac{\partial z_{ki}^2}{\partial \psi_{ki}^2} = g_2'(\psi_{ki}^2), \quad (24)$$

with $g_2(\cdot) = \text{Tanh}$ and $g_2'(\cdot) = 1 - (g_2(\cdot))^2$. Thus, (23) can be expressed as

$$\frac{\partial E}{\partial w_{i, \text{Re}}^2} = \frac{1}{K} \sum_{k=1}^K e_k w_i^{3, \text{Re}} z_k^1 g_2'(\psi_{ki}^2). \quad (25)$$

Similarly, we can also obtain the derivative with respect to $w_i^{2, \text{Im}}$, i.e.,

$$\frac{\partial E}{\partial w_{i, \text{Im}}^2} = \frac{1}{K} \sum_{k=K+1}^{2K} e_k w_i^{3, \text{Im}} z_k^1 g_2'(\psi_{ki}^2). \quad (26)$$

Note that the weight matrix $\mathbf{W}^1$ has a special structure as in (13). For the element belonging to the matrix $\mathbf{W}^{11}$, we have $w_{ij}^1 = w_{(i+K)(j+N)}^1$, with the index $i \in [1, K]$ and $j \in [1, N]$. For the element belonging to the matrix $\mathbf{W}^{12}$, we have $w_{ij}^1 = -w_{(i+K)(j-N)}^1$ with the index $i \in [1, K]$ and $j \in [N+1, 2N]$. Thus, if w_{ij}^1 is one of the elements in $\mathbf{W}^{11}$, the partial derivative with respect to w_{ij}^1 can be computed as

$$\frac{\partial E}{\partial w_{ij}^1} = \frac{1}{2K} \sum_{k \in \{i, i+K\}} \frac{\partial E_k}{\partial z_k^3} \frac{\partial z_k^3}{\partial \psi_k^3} \sum_{n=1}^{N'} \frac{\partial \psi_k^3}{\partial z_{kn}^2} \frac{\partial z_{kn}^2}{\partial \psi_{kn}^2} \frac{\partial \psi_{kn}^2}{\partial z_k^1} \frac{\partial z_k^1}{\partial \psi_k^1} \frac{\partial \psi_k^1}{\partial w_{ij}^1}, \quad (27)$$

where for $k = i$

$$\frac{\partial \psi_k^3}{\partial z_{kn}^2} = w_n^{3, \text{Re}}, \quad \frac{\partial \psi_{kn}^2}{\partial z_k^1} = w_n^{2, \text{Re}}, \quad \frac{\partial \psi_k^1}{\partial w_{ij}^1} = z_j^0, \quad (28)$$

and for $k = i + K$

$$\frac{\partial \psi_k^3}{\partial z_{kn}^2} = w_n^{3, \text{Im}}, \quad \frac{\partial \psi_{kn}^2}{\partial z_k^1} = w_n^{2, \text{Im}}, \quad \frac{\partial \psi_k^1}{\partial w_{ij}^1} = z_{j+N}^0. \quad (29)$$

Substituting (28) and (29) into (27), we have

$$\frac{\partial E}{\partial w_{ij}^1} = \frac{1}{K} \sum_{n=1}^{N'} [e_i w_n^{3, \text{Re}} w_n^{2, \text{Re}} z_j^0 g_2'(\psi_{in}^2) + e_{i+K} w_n^{3, \text{Im}} w_n^{2, \text{Im}} z_{j+N}^0 g_2'(\psi_{(i+K)n}^2)]. \quad (30)$$

Similarly for w_{ij}^1 in $\mathbf{W}^{12}$, the partial derivative

$$\frac{\partial E}{\partial w_{ij}^1} = \frac{1}{K} \sum_{n=1}^{N'} [e_i w_n^{3, \text{Im}} w_n^{2, \text{Im}} z_j^0 g_2'(\psi_{in}^2) - e_{i+K} w_n^{3, \text{Re}} w_n^{2, \text{Re}} z_{j-N}^0 g_2'(\psi_{(i+K)n}^2)]. \quad (31)$$

With the same way, the derivatives on $\{b_i^l\}$ can also be derived.

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